THE SELF-LEARNING UNIVERSE

FROM LEARNING DYNAMICS
TO GAUGE THEORIES AND GRAVITY

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Simulation vs. learning hypothesis

Simulation hypothesis

proposes that what one experiences as the real world ~~is actually a simulated reality~~ can be successfully modeled as a computer simulation in which fields, particles, and humans are different-level constructs.

Learning hypothesis

proposes that what one experiences as the real world ~~is actually a learning system~~ can be successfully modeled as a learning system in which fields, particles, and humans are different-level learning agents.

Sits at the crossing of four threads:

- information-theoretic (Wheeler, Deutsch)
- entropic gravity (Jacobson, Verlinde)
- stochastic dynamics (Nelson)
- geometric learning (Amari)

What is different here: no quantum postulate, no fields, no gravitational postulate, no spacetime.

Dictionary under construction:

agents $\Longleftrightarrow$ **fermions** [Andrejic, VV (2023)]
preconditioner $\Longleftrightarrow$ **metric** [Guskov, VV (2025)]
momentum $\Longleftrightarrow$ **inertia** [Gusev, VV (2025)]
... $\Longleftrightarrow$ ...

Only two knobs in the whole talk

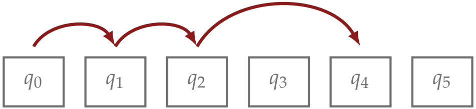
1. How many steps enter the objective.
one $\Rightarrow$ first order (gradient flow)   two $\Rightarrow$ second order (inertia, geodesics)

2. How much the agent can measure.
Δq, S_x, N inaccessible $\Rightarrow$ replace by A_μ, T, μ

Every equation below is one setting of these two knobs.

Time and space-time

Time. N trainable variables $q \in \mathbb{R}^N$, one global objective $\mathcal{S}[q]$, memory for only $d \ll N$ of them.
You must split and process sequentially: $q = (q_0, q_1, \ldots), q_n \in \mathbb{R}^d$ where the block index n *is* time.
Proper time. More generally the agent may skip blocks which is tracked by a dynamical bookkeeping variable q_n^0.



The global objective must be modelled locally:

$$\mathcal{S}[q] \approx \varepsilon \sum_n L_x(q_n, x_n) \xrightarrow[\rho(x_n|q_n)]{\text{integrate out } x} \mathcal{S}[q] \approx \varepsilon \sum_n V(q_n)$$

Zeroth order: the loss depends on *where* the agent is.

The metric is the preconditioner

$g_{\mu\nu}$ enters as the shape of the trust region, as in covariant gradient descent [Guskov, VV (2025)], extended here to include time.

$$q_{n+1} = \arg\operatorname{extr}_q \left\{ \varepsilon V(q) + \frac{1}{2\gamma} g_{\mu\nu}(q_n)\, \Delta q^\mu \Delta q^\nu \right\} \Rightarrow \dot{q}^\mu = -\gamma\, g^{\mu\nu} \partial_\nu V$$

The update rule

Same move, one order up. Let the fast variables remember the previous block too:

$$\mathcal{S}[q] \approx \varepsilon \sum_n L_x(q_n, x_n) \xrightarrow[\rho(x_n|q_n,q_{n-1})]{\text{integrate out } x} \mathcal{S}[q] \approx \varepsilon \sum_k L(q_k, \Delta q_k), \qquad L(q_{n-1}, \Delta q_{n-1}) \equiv \langle L_x(q_n, x_n) \rangle$$

That extra argument Δq is the story of the next three slides. The agent varies the block it occupies, holding q_{n-1} known and q_{n+1} unknown:

$$q_n = \arg\operatorname{extr}_q \left\{ \varepsilon \big(L(q_{n-1}, q - q_{n-1}) + L(q, q_{n+1} - q) \big) + \tfrac{1}{2\gamma} g_{\mu\nu}(q_{n-1})(q - q_{n-1})^\mu (q - q_{n-1})^\nu \right\}$$

Against the *exact* stationarity of $\mathcal{S}$, the local update rule differs by one term:

$$\underbrace{\partial_{q_n^\mu} L - \partial_{\Delta q_n^\mu} L + \partial_{\Delta q_{n-1}^\mu} L}_{\text{discrete Euler–Lagrange}} + \underbrace{\frac{g_{\mu\nu}(q_{n-1})}{\gamma} \frac{q_n^\nu - q_{n-1}^\nu}{\varepsilon}}_{\text{the penalty}} = 0$$

A warning about signatures

With Lorentzian $g_{\mu\nu}$ the penalty is indefinite: the local rule finds saddles, but they correspond to minima of a global objective.

As $\gamma \to \infty$ the penalty drops out, recovering exact Euler–Lagrange.

Physics from learning

The exact step loss $L(q, \Delta q, S_x, N)$ needs three extensive quantities the agent does not control: a displacement *not yet taken*, an entropy it *cannot count*, a fluctuating *number* of hidden variables.

$$\begin{tikzcd}
L(q, \Delta q, S_x, N) \arrow[d, "\Delta q^\mu \to A_\mu"] & \\
L^{(A)}(q, A, S_x, N) \arrow[d, "S_x \to T"] & \textbf{gauge force} \\
L^{(A,T)}(q, A, T, N) \arrow[d, "N \to \mu"] & \textbf{thermal force} \\
L^{(A,T,\mu)}(q, A, T, \mu) & \textbf{quantum phase}
\end{tikzcd}$$

The familiar thermodynamic ladder

extensive	$\to$	**intensive**	
Δq^μ	$\to$	A_μ	*volume* $\to$ *pressure*
S_x	$\to$	T	*entropy* $\to$ *temperature*
N	$\to$	μ	*number* $\to$ *chem. potential*

The last two are literally thermodynamics. The first is too, with one twist: Δq^μ is a vector, so its conjugate "pressure" A_μ carries an index and can have a curl.

Two steps buy a force term; the third buys quantumness:

$$f_\mu = \partial_\mu L^{(A,T)} + \underbrace{Q' A_\mu}_{\text{gauge}} + \underbrace{kT\,\partial_\mu \log \rho}_{\text{thermal}} \qquad \underbrace{S \sim S - \mu\varepsilon\Delta N}_{\text{quantum phase}}$$

The gauge field

First step. The agent cannot condition the fast variables on a displacement it has not yet taken, so the canonical ensemble is the right description:

$$L^{(A)}(q, A) = L(q, \Delta q) - Q' A_\mu \Delta q^\mu$$

Gauge redundancy. Under $A_\mu \to A_\mu + \partial_\mu \chi$ the modelled step loss changes by

$$Q' \partial_\mu \chi \, \Delta q^\mu \;\approx\; Q' \left[\chi(q+\Delta q) - \chi(q) \right]$$

a total step difference, which telescopes under $\sum_n$ and leaves every local stationarity condition unchanged.

One-step dynamics:

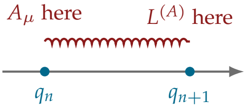
$$\dot{q}^\mu = -\gamma g^{\mu\nu} \left(\partial_\nu L^{(A)} + Q' A_\nu \right)$$

The total force is the gradient of the canonical loss plus the background gauge field, but no electromagnetic force yet.

Electromagnetic tensor

A canonical agent knows only $L^{(A)}(q)$ and $A_\mu(q)$; it must *model* the step loss it cannot see:

$$L(q, \Delta q) \;\approx\; L^{(A)}\big(\underbrace{q + \Delta q}_{\text{where the fast variables \textit{will} equilibrate}}\big) \;+\; Q' A_\mu\big(\underbrace{q}_{\text{all it can read \textit{before} stepping}}\big)\,\Delta q^\mu$$



The punchline

Evaluate both at the *same* point and the Lorentz force vanishes identically. Electromagnetism is the causal asymmetry of the update rule, not an extra field.

With $A_\mu(q_{n-1}) - A_\mu(q_n) \approx -\Delta q^\nu_{n-1}\partial_\nu A_\mu$ and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, the two-step objective gives

$$\dot q^\mu = -\gamma\, g^{\mu\nu}\left(\partial_\nu L^{(A)} + Q'\varepsilon F_{\nu\sigma}\dot q^\sigma\right)$$

$\sigma = 0$ gives the electric force, $\sigma = j$ the magnetic one.

Still first order: the penalty is quadratic in a *single* displacement.

Coarse-graining trajectory

Smooth one agent's history over a time $\tau_q \gg \varepsilon$ and a scale σ in parameter space (exponential window $\times$ Gaussian kernel) $\Rightarrow$ fields $\rho(t,q)$, $v^\mu(t,q)$, with $\int d^d q\, \rho = 1$. Take $\dot{q}^0 = 1$, so $v^0 = 1$, $g_{00} = -1$, $g_{0i} = 0$; the spatial g_{ij} stays curved.

$$v^i = -\gamma g^{ij}\left(\nabla_j L^{(A,T)} + Q' A_j + kT\nabla_j \log \rho\right) \xrightarrow[\text{continuity}]{} \partial_0 \rho = \gamma \nabla_i \left[\rho g^{ij}\left(\nabla_j L^{(A,T)} + Q' A_j\right)\right] + \gamma kT \Delta \rho$$

Stochastic Gradient Descent

Drift + diffusion is the continuous-time picture of stochastic gradient descent. kT is the **gradient-noise temperature**, set by batch size and learning rate.

Hyperparameters

$$m = \frac{\varepsilon}{\gamma}, \qquad \hbar = 2\varepsilon |kT|, \qquad h = |\mu| \varepsilon$$

Irreversible transport from learning

At learning equilibrium, the irreversible transport vanishes

scalar sector: irreversible *entropy production* · spinor sector (later): irreversible *spin transport*

Multiply the velocity equation by $kT\rho\,\nabla_i \log\rho$. The result is an identity that splits the entropy budget in two:

$$\underbrace{k^2T^2\rho\,\gamma g^{ij}\nabla_i \log\rho\,\nabla_j \log\rho}_{\text{irreversible}} = \underbrace{-v^i kT\rho\,\nabla_i \log\rho}_{\text{total}} \underbrace{-\,\gamma g^{ij}(\nabla_i L^{(A,T)} + Q' A_i)\,kT\rho\,\nabla_j \log\rho}_{\text{reversible}}$$

The total change is *already* in the action; to constrain only the irreversible part, the drift piece must be added back.

That single added term cancels the diffusion term of Fokker–Planck exactly, and completes the action.

Equilibrium need only require that the production *integrate* to zero. A local residual $\rho\,U(q)/\varepsilon$ survives: $U(q)$ is not fixed by coarse-graining; later becomes the mass term.

Emergent quantumness

Integrate out v^i and rescale, $S = -\varepsilon L^{(A,T)}$, $Q = -\varepsilon Q'$:

$$\mathcal{A}[\rho, S] = \int dt\, d^d q \sqrt{g} \Big[\tfrac{\rho}{\varepsilon} \big(\partial_0 S + Q A_0 + \tfrac{\gamma}{2\varepsilon} g^{ij} (\nabla_i S + Q A_i)(\nabla_j S + Q A_j) + U \big) + 2\gamma k^2 T^2 g^{ij} \nabla_i \sqrt{\rho}\, \nabla_j \sqrt{\rho} \Big]$$

Vary $\sqrt{\rho} \Rightarrow$ quantum Hamilton–Jacobi; vary $S \Rightarrow$ continuity. With $m = \varepsilon / \gamma$ and $\hbar = 2\varepsilon |kT|$:

$$(\partial_0 S + Q A_0) + \tfrac{1}{2m} g^{ij} (\nabla_i S + Q A_i)(\nabla_j S + Q A_j) - \tfrac{\hbar^2}{2m} \frac{\Delta_g \sqrt{\rho}}{\sqrt{\rho}} + U = 0$$

$$\partial_0 \rho + \tfrac{1}{m} \nabla_i \big(\rho\, g^{ij} (\nabla_j S + Q A_j) \big) = 0$$

The quantum potential is not inserted; it is the coarse-grained diffusion left over after the equilibrium constraint.

The Schrödinger Equation:

$$i\hbar D_0 \Psi + \tfrac{\hbar^2}{2m} g^{ij} D_i D_j \Psi$$

where $\Psi = \sqrt{\rho}\, e^{iS/\hbar}$ and $D_\mu = \nabla_\mu + i\frac{Q}{\hbar} A_\mu$. Its imaginary part of is continuity, the real part is Hamilton–Jacobi.

The Planck constant

The last Legendre transform trades the *number* N of non-trainable degrees of freedom for the chemical potential μ. But N is an integer, so the grand potential (and with it S) is defined only up to [Katsnelson, VV (2021)]

$$S \ \sim \ S - \mu\varepsilon\, \Delta N, \qquad \Delta N \in \mathbb{Z}.$$

Identifying the quantum of action with the chemical potential scale,

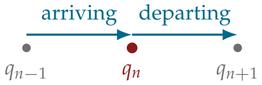
$$h = 2\pi\hbar = 4\pi\varepsilon|kT| = |\mu|\varepsilon,$$

the wavefunction $\Psi = \sqrt{\rho}\, e^{iS/\hbar}$ is automatically single-valued, since $e^{ih\Delta N/\hbar} = e^{i2\pi\Delta N} = 1$. The sign of μ is absorbed into ΔN.

Geodesic equation

One penalty ⇒ gradient descent: the agent has *no memory of how fast it was moving*. That is Aristotle, not Newton. Penalise the departing step as well as the arriving one:

$$q_n = \arg\operatorname{extr}_q \Big\{ \varepsilon\big(L(q_{n-1}, q - q_{n-1}) + L(q, q_{n+1} - q)\big) + \tfrac{1}{2\gamma} g_{\mu\nu}(q_{n-1})(q - q_{n-1})^2 + \tfrac{1}{2\gamma} g_{\mu\nu}(q)(q_{n+1} - q)^2 \Big\}$$



The arriving penalty has its metric at the *fixed* point q_{n-1}, the departing one at the *varied* point q_n. With $p_\mu^{(n)} = \frac{1}{\gamma} g_{\mu\nu}(q_n)\Delta q_n^\nu$ the two contribute

$$p_\mu^{(n-1)} - p_\mu^{(n)} + \tfrac{1}{2\gamma}(\partial_\mu g_{\alpha\beta})\Delta q^\alpha \Delta q^\beta$$

a difference of momenta plus a metric derivative. The Christoffel symbols assemble themselves:

$$\ddot{q}^\mu + \Gamma^\mu_{\nu\rho}\dot{q}^\nu \dot{q}^\rho = \tfrac{1}{m} g^{\mu\nu}\big(\partial_\nu L^{(A)} + Q'\varepsilon F_{\nu\rho}\dot{q}^\rho\big)$$

Momentum SGD on a curved manifold *is* a charged particle in curved spacetime.

Proper length

The quadratic penalty is not the only natural choice. Replace both by the reparametrisation-invariant proper length of the step,

$$-m\sqrt{-g_{\mu\nu}(q_n)\Delta q_n^{\mu}\Delta q_n^{\nu}}.$$

The inertial structure survives intact: the difference of momenta becomes a covariant derivative along the trajectory,

$$\frac{p_\mu^{(n)} - p_\mu^{(n-1)}}{\varepsilon} \;\to\; \frac{D}{d\tau}\left(\frac{m\dot{q}_\mu}{\sqrt{-\dot{q}^2}}\right),$$

with the relativistic conjugate momentum $p_\mu = m\, g_{\mu\nu}\dot{q}^\nu/\sqrt{-g_{\alpha\beta}\dot{q}^\alpha\dot{q}^\beta}$ satisfying the mass shell *automatically*:

$$g^{\mu\nu}p_\mu p_\nu = -m^2, \qquad \frac{D\dot{q}^\mu}{d\tau} = \tfrac{1}{m}g^{\mu\nu}\left(\partial_\nu L^{(A)} + Q'\varepsilon F_{\nu\rho}\dot{q}^\rho\right)$$

Free trajectories are geodesics traced in proper time, $g_{\mu\nu}\dot{q}^\mu\dot{q}^\nu = -1$.
Consistency check. For $|\Delta q^i| \ll |\Delta q^0| \approx \varepsilon$,

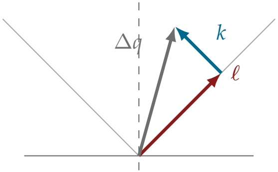
$$-m\sqrt{-g_{\mu\nu}\Delta q^\mu \Delta q^\nu} \approx -m\varepsilon + \tfrac{1}{2\gamma}g_{ij}\Delta q^i \Delta q^j + \dots$$

The constant drops out of the variation and what remains is *exactly* the quadratic penalty used so far.

Spinors (or step coordinates)

The block structure fixes a preferred direction $t^\mu = (1,0,0,0)$. A timelike step then splits uniquely into two future-pointing null vectors:

$$\Delta q_n^\mu = \ell_n^\mu + k_n^\mu, \quad \ell, k = \frac{\Delta q^0 \pm \sqrt{g_{ij}\Delta q^i \Delta q^j}}{2}\,(t^\mu \pm \hat{n}^\mu), \quad \ell^2 = k^2 = 0$$



In 3+1 dimensions a future-pointing null vector is a spinor bilinear. Introduce two chiral projections ξ (left) and η (right) in the Weyl basis:

$$\ell^\mu = \bar{\xi}\gamma^\mu\xi, \qquad k^\mu = \bar{\eta}\gamma^\mu\eta, \qquad \Delta q^\mu = \bar{\xi}\gamma^\mu\xi + \bar{\eta}\gamma^\mu\eta$$

What this says

Chirality is not added to the theory. It *is* the two-fold splitting of a single learning step. A spinor is a square root of a null direction; left and right are the two halves of the step.

Mass in the step-length penalty

Fierz identity for a left- and a right-handed spinor: $g_{\mu\nu}(\bar{\xi}\gamma^\mu\xi)(\bar{\eta}\gamma^\nu\eta) = -2|\bar{\xi}\eta|^2$, so the squared displacement reduces to the *cross term* between the two null halves, $-g_{\mu\nu}\Delta q^\mu \Delta q^\nu = 4|\bar{\xi}\eta|^2$, and hence

$$\underbrace{m\sqrt{-g_{\mu\nu}\Delta q^\mu \Delta q^\nu}}_{\text{penalty on how far you move}} = 2m|\bar{\xi}\eta| = \underbrace{m(\bar{\xi}\eta + \bar{\eta}\xi)}_{\text{Dirac mass term}}$$

Read it in both directions. A step of zero length costs nothing and couples nothing: a massless field has decoupled chiralities. A massive field is one whose learning steps are *penalised by their length*, and that penalty is exactly what ties left to right.

Mass

m is not a property of a particle here. It is the exchange rate between how far the learner moves and how much that costs, already fixed as $m = \varepsilon/\gamma$, the ratio of time step to learning rate.

Varying the spinors

Every equation so far came from varying the *block* variable q_n. The spinors live on *steps*, so varying them gives a complementary equation from the same objective:

$$(\xi_n, \eta_n) = \arg\operatorname{extr}_{\xi,\eta}\Big\{ -S(q_n + \Delta q) - QA_\mu(q_n)\Delta q^\mu - m(\bar{\xi}\eta + \bar{\eta}\xi) \Big\}$$

Treat $\xi, \bar{\xi}$ as independent and use $\partial\Delta q^\mu/\partial\bar{\xi} = \gamma^\mu\xi$:

$$\gamma^\mu(\partial_\mu S + QA_\mu)\xi + m\,\eta = 0$$

$$\gamma^\mu(\partial_\mu S + QA_\mu)\eta + m\,\xi = 0$$

The two chiralities are coupled *only* through m. Forming $\psi^\pm = \xi \pm \eta$ decouples them:

$$\gamma^\mu(\partial_\mu S + QA_\mu)\psi^\pm \pm m\,\psi^\pm = 0$$

The two are related by a field redefinition; take ψ^+.

Two variations, two kinds of law

Vary the block $\Rightarrow$ geodesic equation: *where* the agent goes.

Vary the step $\Rightarrow$ spinor equation: *how* the step is internally structured.

Dirac equation

Irreversible spin transport

$$\int d^4q\sqrt{-g}\,\rho\,\mathrm{Im}\big(\bar{\chi}\gamma^\mu\mathcal{D}_\mu\chi\big) = 0$$

Why the imaginary part? Because $\mathrm{Re}(\bar{\chi}\gamma^\mu\mathcal{D}_\mu\chi) = \frac{1}{2}\nabla_\mu(\bar{\chi}\gamma^\mu\chi)$ is *already* fixed by continuity. The imaginary part is the independent degree of freedom, and it is the spinor incarnation of the same equilibrium condition that gave Schrödinger.

Adding this constraint to the algebraic action and simplifying (backup) turns the pointwise relation into a wave equation:

Dirac on a curved background

$$\Big(i\hbar\gamma^\mu\big(\partial_\mu + \tfrac{1}{4}\omega_\mu^{ab}\sigma_{ab} + i\tfrac{Q}{\hbar}A_\mu\big) - m\Big)\psi = 0$$

One postulate, two sectors:
entropy production $\to$ Schrödinger; *spin transport* $\to$ Dirac.

Parallel learning

One agent traces a one-dimensional trajectory. Deploy many, but parallelism without coordination merely multiplies independent optimizations.

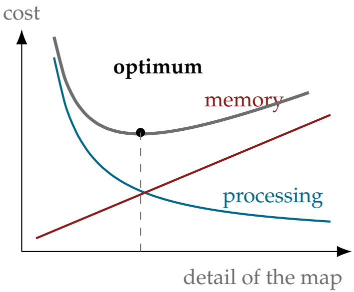
The algorithmic remedy is *locality of visitation*: agents arriving at nearby blocks tend to remain neighbours at the next step. This statistical regularity partitions the trainable space into cells.

	Agents	**Cells**
variables	trainable	non-trainable (fast)
role	**processing**	**memory**
carries	state ψ, position q	$g_{\mu\nu}$, A_μ, ω_μ^{ab}
dynamics	hopping, updating	thermodynamic relaxation
becomes	Dirac equation	Einstein–Maxwell

For the systems people

A cell is a **local cache of agent activity**, a parameter server shard holding intensive state, with all the familiar pathologies: staleness, synchronization cost, lossy compression.

The trade-off that fixes gravity



Two competing overheads.

- **Processing efficiency**: the algorithmic effort of advancing an agent's internal state. A poor or inconsistent map means algorithmic friction: agents explore inefficiently.
- **Memory cost**: the informational price of storing and synchronising agent activity. Track every fluctuation and you get a memory explosion; compress to a flat featureless map and you precondition nothing.

The field equations are the algorithmic fixed point where these two costs balance.

Einstein and Maxwell equations

What can the cell's free energy be? Three requirements, all inherited from the construction: locality (the potential density depends on local values and derivatives only) · diffeomorphism invariance (cell labels are arbitrary) · gauge invariance (inherited from the step loss). Landau theory logic, with no microscopic model of the fast variables required:

$$\mathcal{F} = \int d^4q\sqrt{g}\Big[\underbrace{\tfrac{2\Lambda}{16\pi G}}_{\text{order }0} \underbrace{- \tfrac{R}{16\pi G} + \tfrac{1}{4e^2}F^2}_{\text{leading derivatives}} + \ldots + \bar{\Psi}\big(\hbar\gamma^\mu\mathcal{D}_\mu - m\big)\Psi\Big]$$

Read the terms as an accountant would

- Λ: the irreducible memory cost of maintaining the map at all, even with all agent activity coarse-grained away. *The price of the network's own existence.*
- R: the extra bits curvature costs over a flat map.
- $F_{\mu\nu}F^{\mu\nu}$: the cost of a non-trivial gauge configuration.

A hierarchy without tuning

G and e *run*. Coarse-graining stiffens the metric, so G^{-1} grows while Λ stays the baseline overhead: $\Lambda \ll G^{-1}$ reflects the number of scales integrated over, not tuning.

The learning algorithm

Because the fast variables are in local equilibrium, the cell potential is an effective action for the slow fields, and $\delta\mathcal{F} = 0$ becomes the principle of stationary action:

Einstein–Maxwell–Dirac

$$\mathcal{A} = \int d^4q\sqrt{-g}\left[\frac{1}{16\pi G}(R - 2\Lambda) - \frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}\left(i\hbar\gamma^\mu\left(\mathcal{D}_\mu + i\tfrac{Q}{\hbar}A_\mu\right) - m\right)\Psi\right]$$

Processor

The **Dirac** term is the *update rule for agents*: how an agent's internal state Ψ evolves using the stored geometric and gauge structure.

Memory

The **Einstein** and **Maxwell** terms are the *update rules for cells*: how the network reorganises $g_{\mu\nu}, A_\mu$ to represent collective agent activity efficiently.

Old puzzles

Reframings offered by the framework

- Equivalence principle: a macroscopic observer cannot tell whether the force on microscopic agents is a genuine loss gradient or the collective memory stored in the metric.
- Gauge principle: the update cannot depend on how the system labels its internal displacements, so only gauge-invariant quantities matter.
- Quantum indeterminacy: not fundamental randomness, but the irreducible uncertainty of a coarse-grained description. The wavefunction is a computational tool for representing incomplete knowledge.
- Measurement: apparent collapse is the *refinement* of the coarse-grained description when new information arrives.

What is actually assumed

- the equilibrium postulate (vanishing irreversible transport)
- the mean-field ansatz $S_x = k \log \rho$
- $d+1 = 4$ inherited from the null/spinor decomposition
- locality of visitation for the cellular partition
- Lorentzian $\leftrightarrow$ Euclidean matching by Wick rotation

Everything else follows from the optimisation alone.

A one-line summary

The Self-Learning Universe

Not a machine executing a program, nor a computation floating in the abstract, but a learning system whose entire existence is the unending task of minimising its own loss.

The dictionary

Learning	Physics	Where it came from
block index n	time t	sequential processing under $d \ll N$
preconditioner	metric $g_{\mu\nu}$	shape of the step penalty
conjugate of displacement	gauge field A_μ	Legendre transform $\Delta q \to A$
causal asymmetry of the update	Lorentz force, $F_{\mu\nu}$	$L^{(A)}$ at $q{+}\Delta q$, A at q
integer count of hidden d.o.f.	quantum phase S	$S \sim S - \mu\varepsilon\Delta N$
momentum	inertia, $\Gamma^{\mu}_{\nu\rho}$	penalising the departing step
step-length penalty	mass $m = \varepsilon/\gamma$	Fierz: $m\sqrt{-\Delta q^2} = m\bar{\xi}\eta + \text{h.c.}$
two null halves of a step	spinors $\psi^{\pm} = \xi \pm \eta$	unique null decomposition
local cache of agent activity	spacetime, $g_{\mu\nu}(q)$, $A_\mu(q)$	locality of visitation
baseline cost of the infrastructure	cosmological constant, Λ	zeroth order of the cell potential
storage cost of a curved map	Einstein-Hilbert action, $R/16\pi G$	leading derivative term